Electrochemical cells

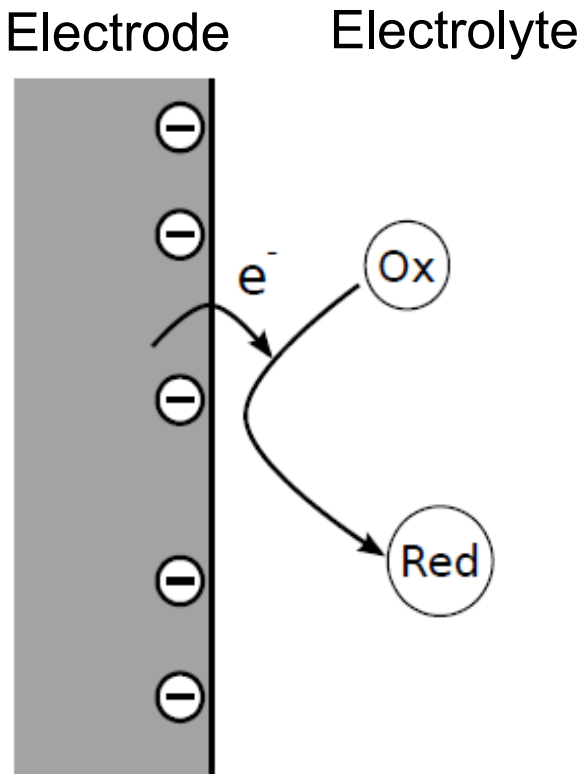
Jens Oluf Jensen

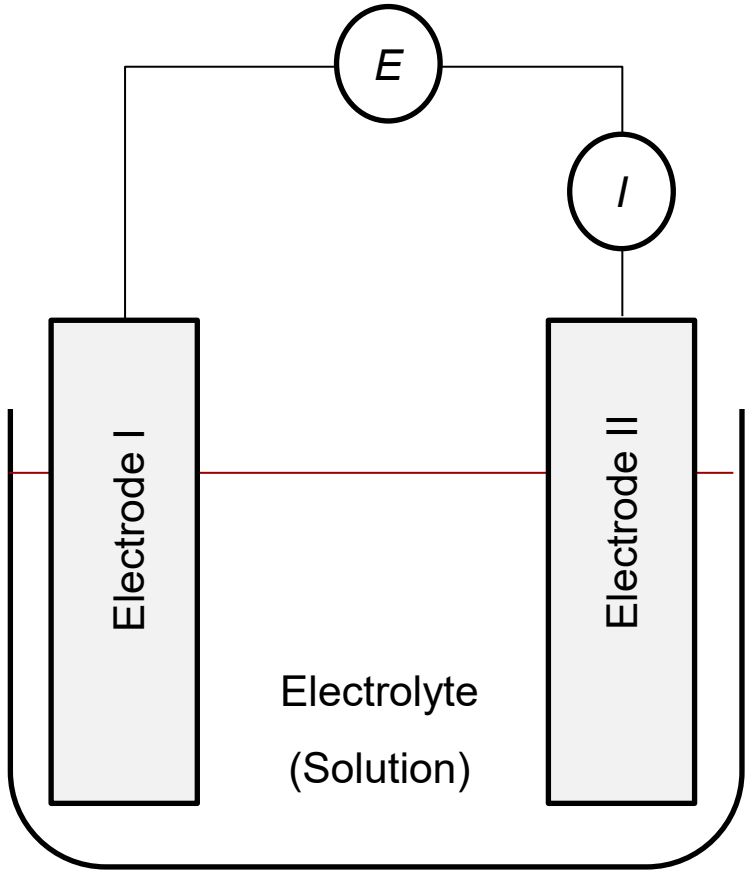
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Outline

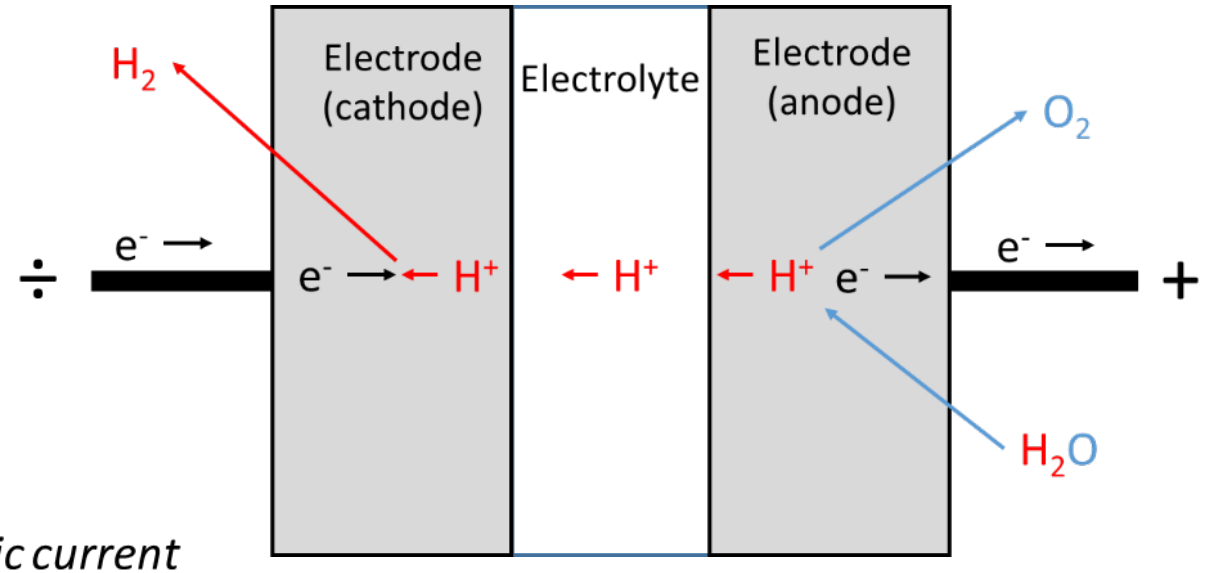
- Electrode and cell potential
- Potential, current and energy
- Some definitions
- Electrochemical reactions
- From thermodynamics to electrochemistry
- Faraday's law
- Activity and standard potentials
- The Nernst equation
- Electrokinetics
- Efficiency

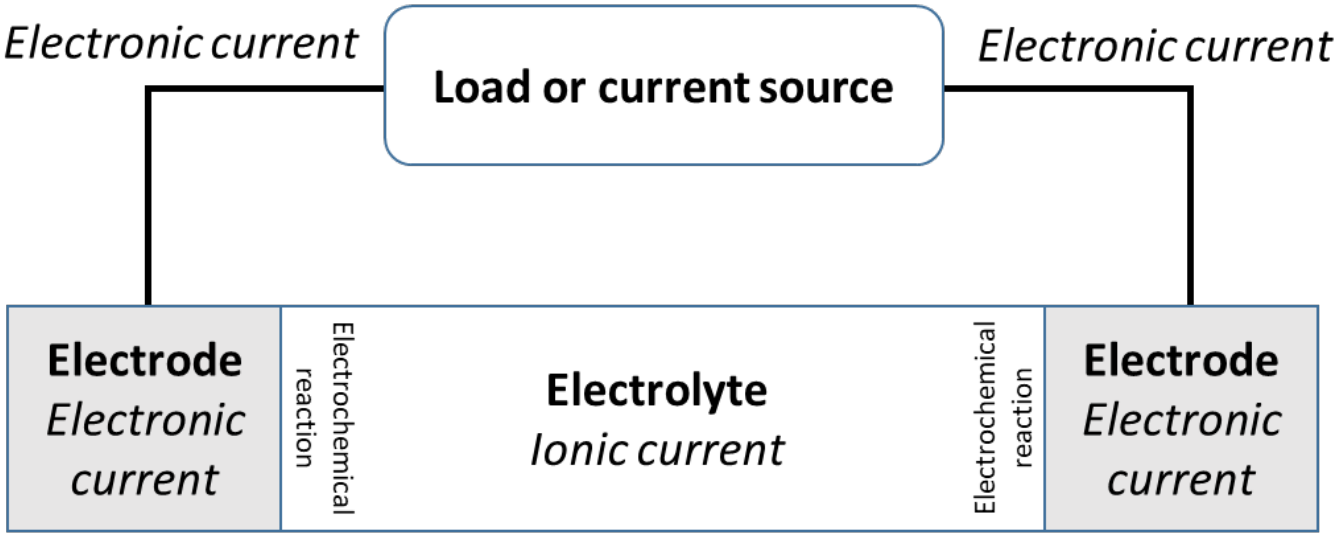
Electrochemical cells





Electrochemical cells

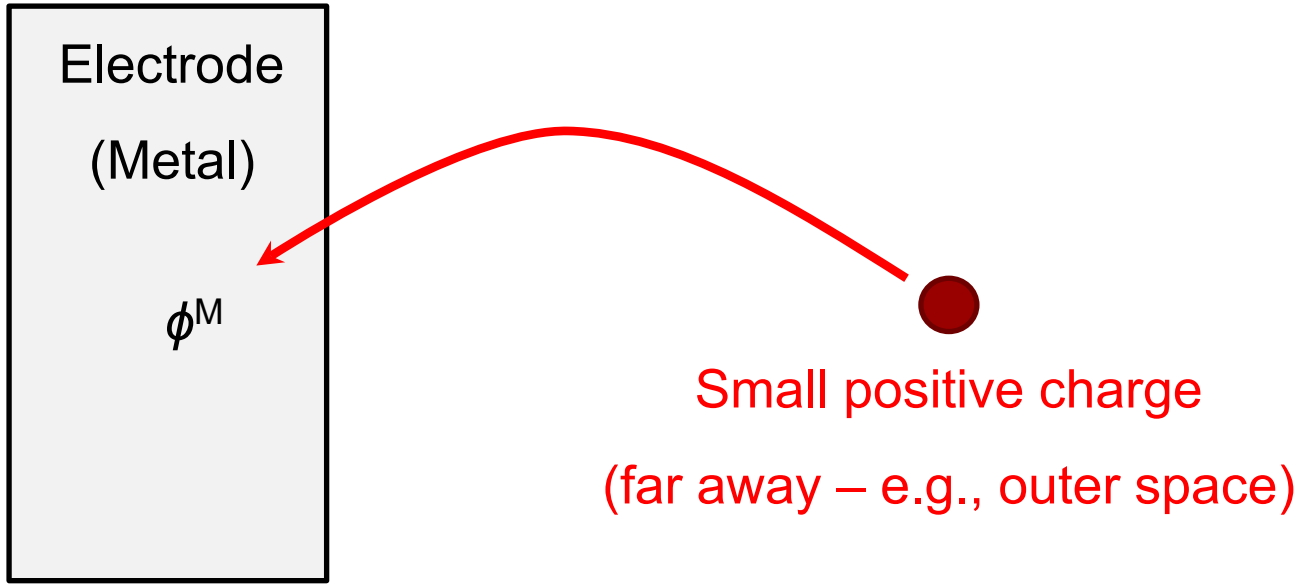




- Batteries
- Fuel cells
- Electrolyzer cells

Electrode potentials (absolute)

ϕ = potential (V)

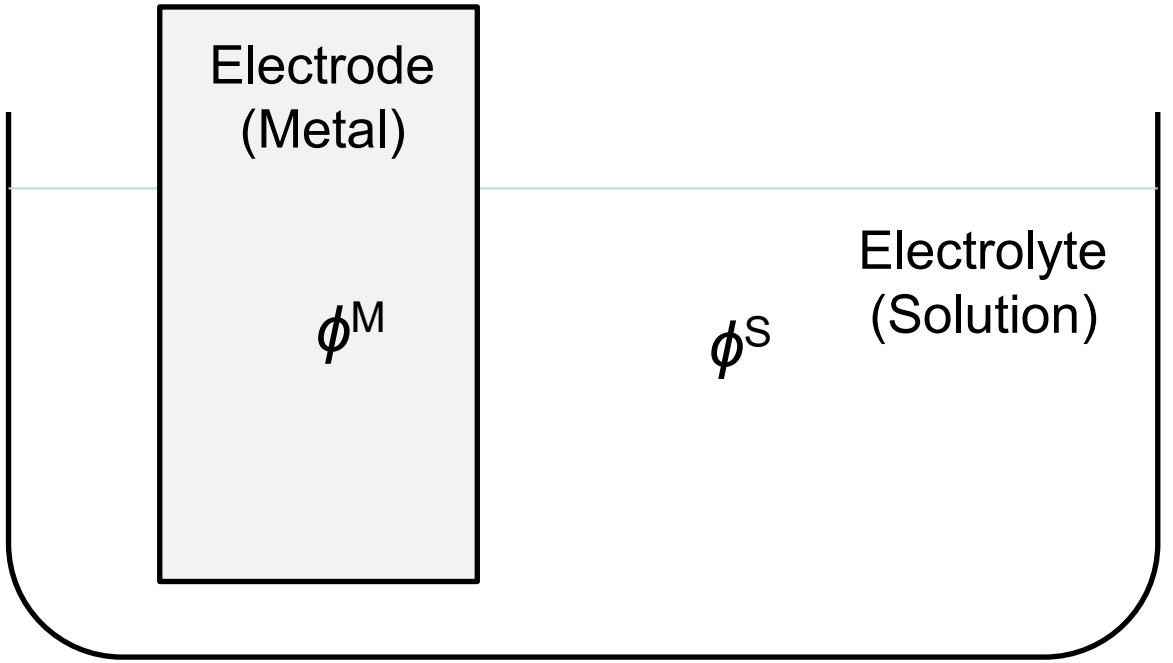


Absolute electrode potential:

$$\phi^{\mathrm{M}} = \frac{\text{The work needed to bring the charge into the electrode}}{\text{The small charge}} = \frac{W}{Q}$$

Electrode potential - a half-cell

ϕ = potential



ϕ^M

$\Delta\phi = \phi^M - \phi^S$

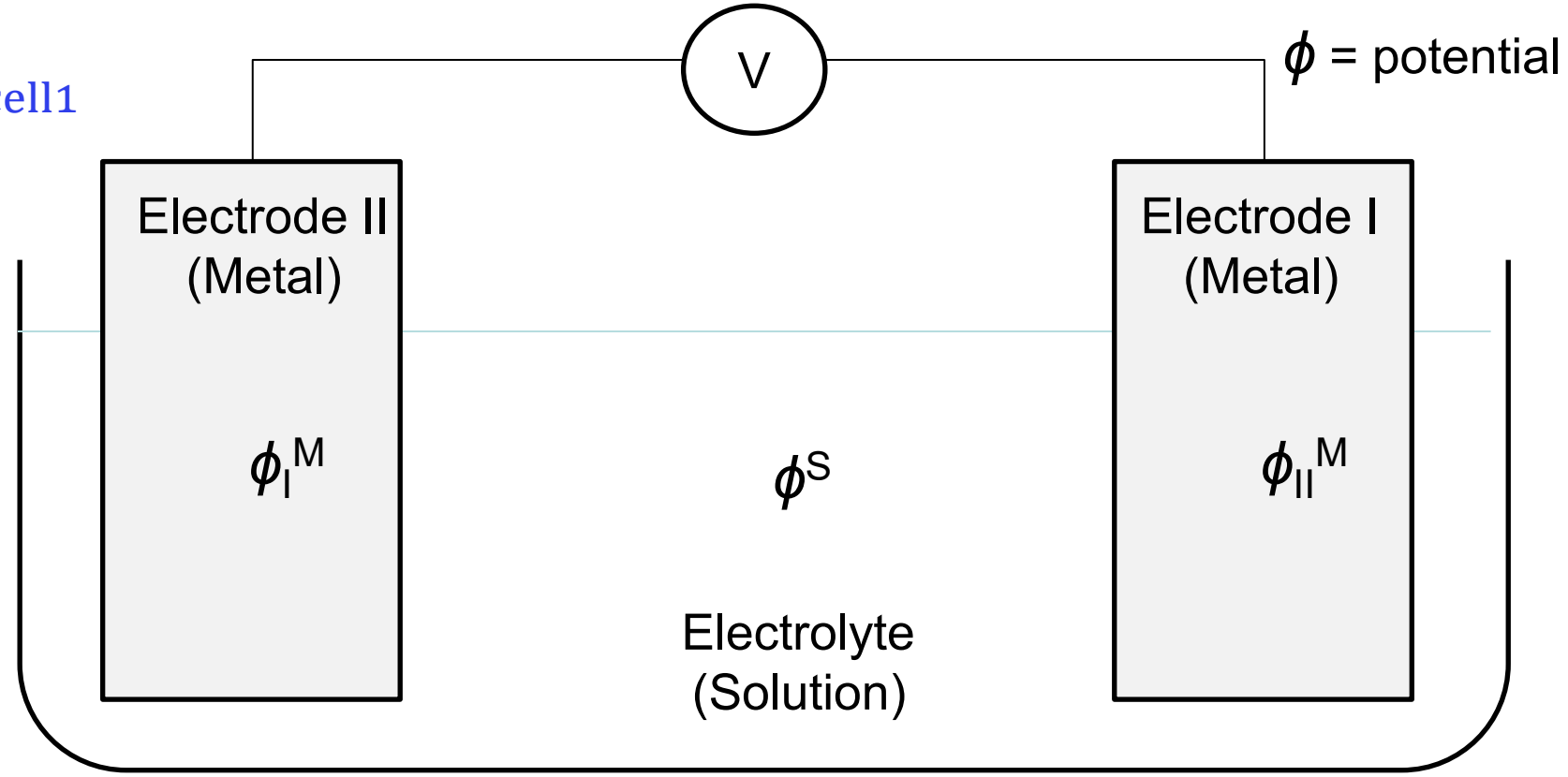
ϕ^S

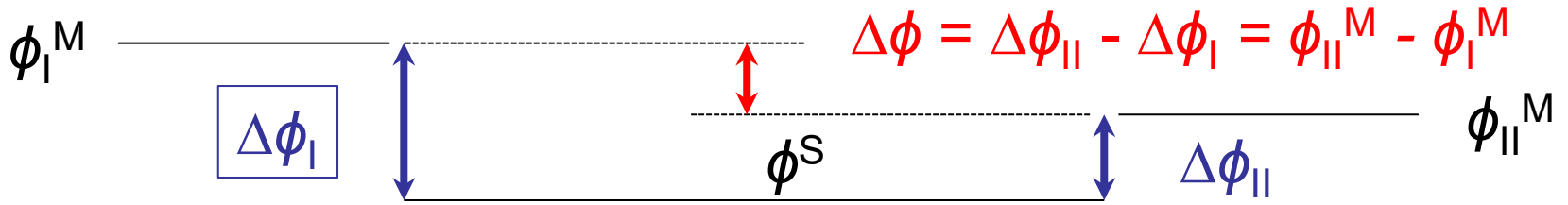
Cell potential

$$E_{\text{cell}} = E_{\text{halfcell2}} - E_{\text{halfcell1}}$$

What is $E_{\text{halfcell1}}$?

What is $E_{\text{halfcell2}}$?





Potential, current and energy

The work delivered or consumed is charge times potential difference:

$$W = Q \cdot \Delta E$$

Charge is:

$$Q = I \cdot t$$

Energy (work):

$$W = I \cdot \Delta E \cdot t$$

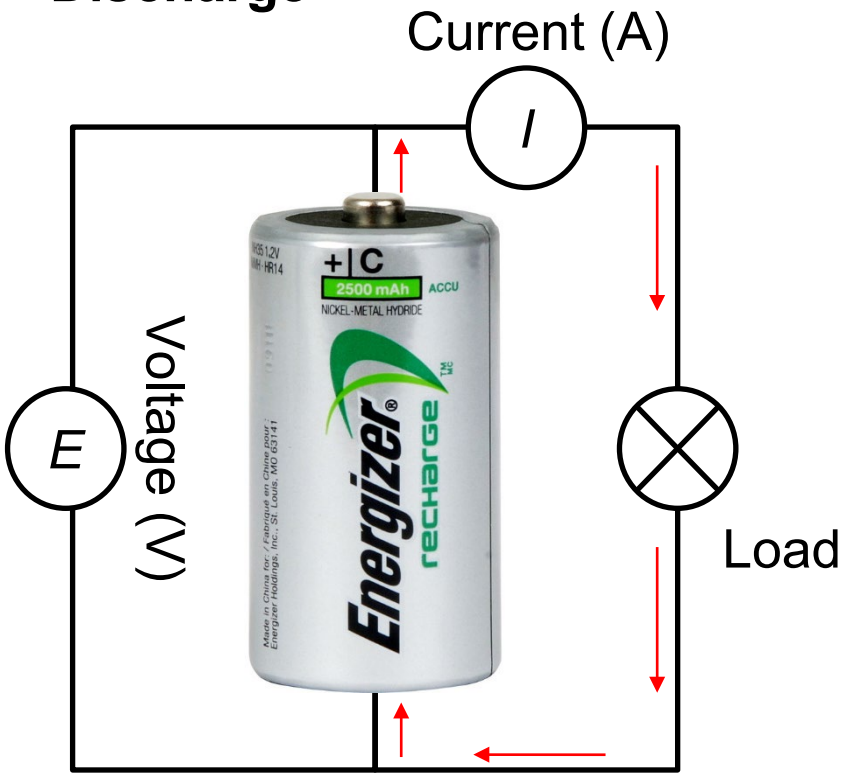
Power is:

$$P = I \cdot \Delta E$$

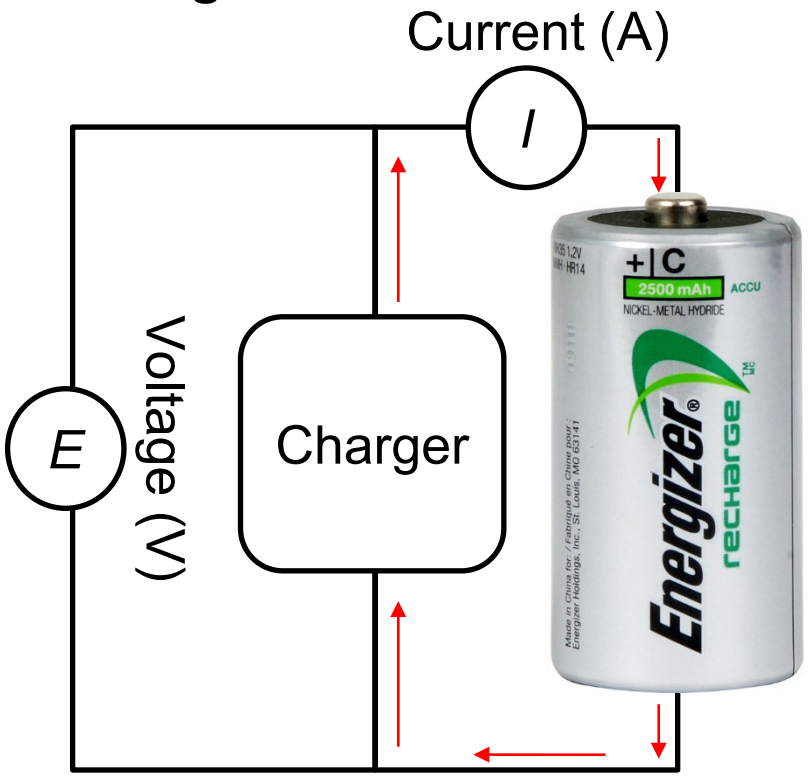
Ohmic voltage drop:

$$\Delta E = R \cdot I$$

Discharge



Charge



W = work (J), Q = charge (C), E = voltage (V), I = current (A), P = power (W)

Batteries vs. fuel cells vs. electrolyzers

	Galvanic cells A chemical process generates an electrical current	**Electrolyzer cells** An electrical current generates a chemical process
Internal reactants and products	Battery (during discharging)	Battery (during charging)
External reactants and products	Fuel cell	Electrolyzer

Definitions – anode and cathode

The **anode** is the electrode where **oxidation** takes place

An anodic process is an oxidation process.

Something is oxidized at the anode and

Electrons are **liberated** at the **anode**

$Fe^{2+} \rightarrow Fe^{3+} + e^-$ (anodic)

$H_2 \rightarrow 2H^+ + 2e^-$ (anodic)

The **cathode** is the electrode where **reduction** takes place

A cathodic process is a reduction process.

Something is reduced at the cathode

Electrons are **consumed** at the **cathode**

$Fe^{3+} + e^- \rightarrow Fe^{2+}$ (cathodic)

$2H^+ + 2e^- \rightarrow H_2$ (cathodic)

Electrochemical reactions

Balancing chemical reactions:

- Same number of each atom on each side (mass balance)
- Same total charge on each side (charge balance)

Balancing electrochemical reactions:

- Same rules
- Electrons, e^-, included

$$Cu \rightleftarrows Cu^{2+} + 2e^-$$

$$Fe^{3+} + e^- \rightleftarrows Fe^{2+}$$

$$CH_3OH + H_2O \rightleftarrows CO_2 + H^+$$



$$CH_3OH + H_2O \rightleftarrows CO_2 + 6H^+ + 6e^-$$

$$H_2 \rightleftarrows 2H^+ + 2e^-$$

Electrochemical reactions (lead-acid battery)

Anodic half-cell reaction: $PbSO_4 + 2H_2O \rightarrow PbO_2 + 4H^+ + SO_4^{2-} + 2e^-$

Cathodic half-cell reaction: $PbSO_4 + 2e^- \rightarrow Pb + SO_4^{2-}$

Total cell reaction: $2PbSO_4 + 2H_2O \rightarrow PbO_2 + Pb + 4H^+ + 2SO_4^{2-}$

From thermodynamics to electrochemistry

From thermodynamics to electrochemistry

Maximum work obtainable from a process

$$w_{\max} = -\Delta g_{\mathrm{r}}$$

Maximum work from movement of charge

$$w_{\max} = q\Delta E$$

$$qE_{\mathrm{cell}} = -\Delta g_{\mathrm{r}}$$

$$q = nF$$

$$E_{\mathrm{cell}} = \frac{-\Delta g_{\mathrm{r}}}{nF}$$

q = charge (not heat), n = number of electrons transferred, F = Faraday's constant (the charge of one mole electrons or protons)

From thermodynamics to electrochemistry

The signing problem

$2PbSO_4 + 2H_2O \rightleftarrows PbO_2 + Pb + 4H^+ + 2SO_4^{2-}$ Charging, positive Δg_r (electrolysis cell)

$PbO_2 + Pb + 4H^+ + 2SO_4^{2-} \rightleftarrows 2PbSO_4 + 2H_2O$ Discharging, negative Δg_r (galvanic cell)

$$E_{cell} = \frac{-\Delta g_r}{nF} \text{ (galvanic cell)}$$

$$E_{cell} = \frac{\Delta g_r}{nF} \text{ (electrolyzer cell)}$$

Faradays law (of electrolysis)

The amount of matter converted
is proportional to
the charge passed through the electrode or the cell.

$$n_{\mathrm{con}} = \frac{Q}{nF} \qquad m_{\mathrm{con}} = \frac{MQ}{nF} \qquad \dot{n}_{\mathrm{con}} = \frac{I}{nF}$$

n_{con}: number of moles species converted

m_{con}: mass of species converted

$\dot{n}_{\mathrm{con}}$: Molar flow of converted species

Q: number of moles elementary charges

M: molar mass of the converted species

nF: charge transferred for the conversion of one mole substance.

Activity, *a* (dimension-less)

A dimension-less alternative to partial pressure or cocentration that makes thermodynamics fit.

Species	Characterized by	Activity	Note
Pure gas	Pressure, p	$\frac{p}{p^0}$	p^0 = 1 bar
Gas component, i, in gas mixture	Partial pressure, p_i	$\frac{p_i}{p^0}$ (approximated)	p^0 = 1 bar
Pure liquid	-	1	
Species, i, dissolved in liquid	Concentration, c_i	$\frac{c_i}{c^0}$ (approximated)	c^0 = 1 M (mol L^{-1})
Pure solid	-	1	

Standard potentials, E^0, at 25 ºC

Half cell potentials cannot be measured. Instead, the standard potential for hydrogen in acidic solution (pH = 0) is used as zero. Other potentials are then measured against hydrogen. All species at activity = 1. I.e., (with approximation) liquids as pure phases, gasses at 1 bar (or 1 atm), ions at 1 molar concertation.

Reaction	E^o (V)	Reaction	E^o (V)
$Li^+ + e^- \rightleftarrows Li$	-3.01	$2H^+ + 2e^- \rightleftarrows H_2$	≡ 0.0000
$Ca^{2+} + 2e^- \rightleftarrows Ca$	-2.84	$Cu^{2+} + 2e^- \rightleftarrows Cu$	0.34
$Na^+ + e^- \rightleftarrows Na$	-2.71	$Cu^+ + e^- \rightleftarrows Cu$	0.52
$Mg^{2+} + 2e^- \rightleftarrows Mg$	-2.84	$Hg^{2+} + 2e^- \rightleftarrows Hg$	0.80
$Al^{3+} + 3e^- \rightleftarrows Al$	-1.66	$Ag^+ + e^- \rightleftarrows Ag$	0.80
$Zn^{2+} + 2e^- \rightleftarrows Zn$	-0.76	$Br_2 + 2e^- \rightleftarrows 2Br^-$	1.07
$Fe^{2+} + 2e^- \rightleftarrows Fe$	-0.44	$O_2 + 4H^+ + 4e^- \rightleftarrows 2H_2O$	1.23
$Cd^{2+} + 2e^- \rightleftarrows Cd$	-0.40	$Cl_2 + 2e^- \rightleftarrows 2Cl^-$	1.36
$Ni^{2+} + 2e^- \rightleftarrows Ni$	-0.23	$F_2 + 2e^- \rightleftarrows 2F^-$	2.87

Examples – complete cell potentials

(standard potentials)

A battery made from zinc and silver

$Ag^+ + e^- \rightleftarrows Ag$	0.80 V
$Zn^{2+} + 2e^- \rightleftarrows Zn$	- (-0.76 V)
Total	1.56 V

Hydrogen fuel cell or a water electrolyzer:

$O_2 + 4H^+ + 4e^- \rightleftarrows 4H_2O$	1.23 V
$2H^+ + 2e^- \rightleftarrows H_2$	- (0.00 V)
Total	1.23 V

Cell potential

Bottom line

- Absolute potentials - irrelevant (cannot be measured practically)
- Single electrode potentials - cannot be measured
- Single electrode potentials - only defined with respect to hydrogen
- Practically, only potential differences are relevant

The Nernst equation (still thermodynamics)

Variation with pressure and concentration (a ≠ 1)
The Nernst equation

Concept: Standard voltage and a compensation

$$E_{\text{cell}} = E^0_{\text{cell}} - \text{compensation}$$

Try with the fuel cell reaction, $2H_2 + O_2 \rightleftarrows 2H_2O$

$$E_{cell} = E^0_{cell} - \frac{RT}{nF} \ln \frac{a^2_{H2O}}{a_{O2} \cdot a^2_{H2}}$$

General Nernst

$$E_{\text{cell}} = E^0_{\text{cell}} \pm \frac{RT}{nF} \ln \frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}}$$

(+ for positive Δg_r - for negative Δg_r)

Variation with pressure and concentration (a ≠ 1)
The Nernst equation

Again, $2H_2 + O_2 \rightleftarrows 2H_2O$ (fuel cell)

$$E_{cell} = E^0_{cell} - \frac{RT}{nF}\ln\frac{a^2_{H2O}}{a_{O2}\cdot a^2_{H2}}$$

$$E_{cell} = E^0_{cell} - \frac{RT}{nF}\ln\frac{1}{a_{O2}\cdot a^2_{H2}}$$

$$E_{cell} = E^0_{cell} + \frac{RT}{nF}\ln\left(a_{O2}\cdot a^2_{H2}\right)$$

Now, $2H_2O \rightleftarrows 2H_2 + O_2$ (electrolyzer)

$$E_{cell} = E^0_{cell} + \frac{RT}{nF}\ln\frac{a_{O2}\cdot a^2_{H2}}{a^2_{H2O}}$$

$$E_{cell} = E^0_{cell} + \frac{RT}{nF}\ln\frac{a_{O2}\cdot a^2_{H2}}{1}$$

$$E_{cell} = E^0_{cell} + \frac{RT}{nF}\ln\left(a_{O2}\cdot a^2_{H2}\right)$$

If the reaction is written differently …

$$E^0_{cell} = \frac{n\left|\Delta g^0_{f,H2O}\right|}{nF}$$

Again, $2H_2 + O_2 \rightleftarrows 2H_2O$ (fuel cell), $n = 4$

$$E_{cell} = E^0_{cell} - \frac{RT}{4F}\ln\frac{a^2_{H2O}}{a^1_{O2}\cdot a^2_{H2}}$$

$$E^0_{cell} = \frac{4\left|\Delta g^0_{f,H2O}\right|}{4F}$$

Again, $H_2 + ½O_2 \rightleftarrows H_2O$ (fuel cell), $n = 2$

$$E_{cell} = E^0_{cell} - \frac{RT}{2F}\ln\frac{a^1_{H2O}}{a^{½}_{O2}\cdot a^1_{H2}}$$

$$E^0_{cell} = \frac{2\left|\Delta g^0_{f,H2O}\right|}{2F}$$

Again, $4H_2 + 2O_2 \rightleftarrows 4H_2O$ (fuel cell), $n = 8$

$$E_{cell} = E^0_{cell} - \frac{RT}{8F}\ln\frac{a^4_{H2O}}{a^2_{O2}\cdot a^4_{H2}}$$

$$E^0_{cell} = \frac{8\left|\Delta g^0_{f,H2O}\right|}{8F}$$

Variation with temperature

$$E_{\text{cell}} = E^0_{\text{cell}} \pm \frac{RT}{nF} \ln \frac{\prod_{\text{Prod.}} a_{\text{i}}^{\nu_{\text{i}}}}{\prod_{\text{React..}} a_{\text{i}}^{\nu_{\text{i}}}}$$

$$E^0_{cell} = \frac{|\Delta g^0_{\text{r}}(T)|}{nF}$$

Electrokinetics

Thermodynamics versus kinetics

Kinetics and thermodynamics are two different angles on processes

Thermodynamics governs the idealized energy

- Minimum electrical energy to charge a battery
- Minimum electrical energy to run an electrolyzer
- Maximum electrical energy form discharging a battery
- Maximum electrical energy from a fuel cell

Kinetics governs the real power

- Real power to charge a battery
- Real power to run an electrolyzer
- Real power from discharging a battery
- Real power from a fuel cell

Polarization and overvoltage

$$A + B \rightleftarrows C + D$$

If the potential of an electrode is different from its equilibrium potential ⇒ it is *polarized*

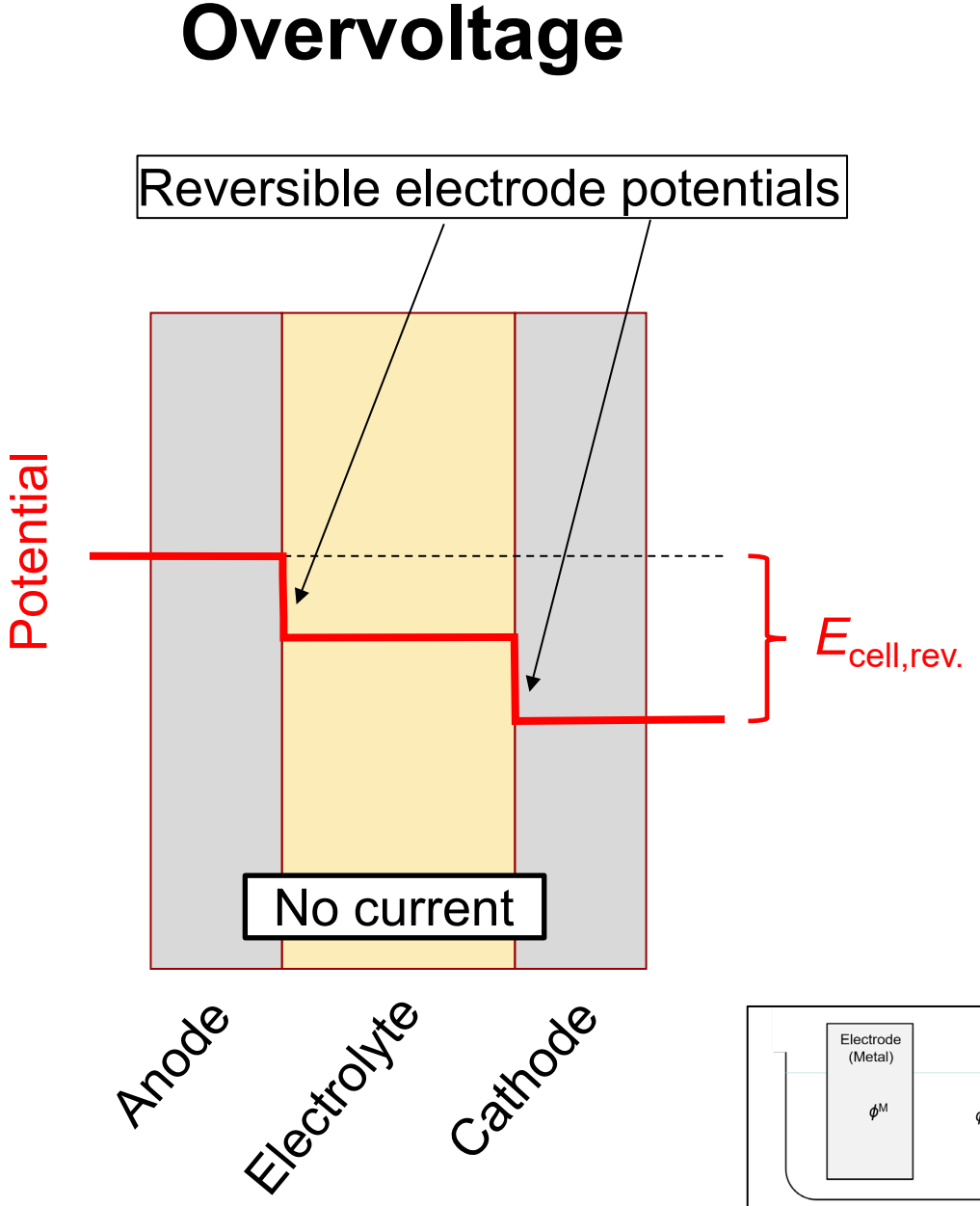
At equilibrium, there is no driving force promoting the reaction in one direction

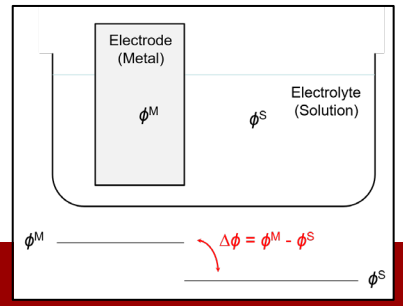
Polarization (voltage) is a driving force for the reaction in one direction (forward or backward)

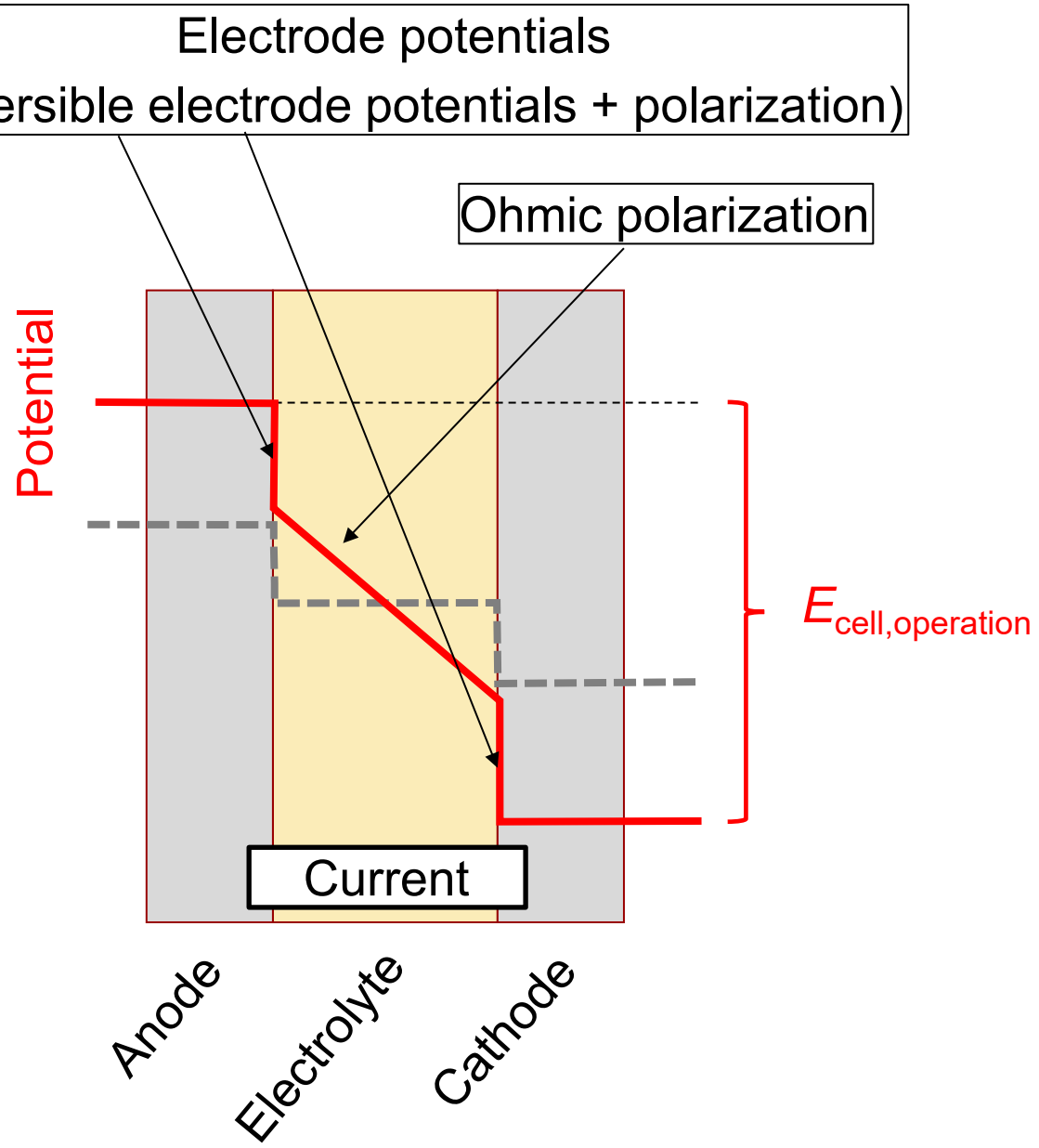
The increase (or decrease) potential is an *overpotential* or *overvoltage*

Overvoltage

(Slightly simplified)







Overvoltage

Sources for overvoltage

- Electrode polarization
- Ohmic drop due to ohmic resistance
- Transport overvoltage (high currents). Reactant concentration lowered near electrodes

Efficiency

$$\eta = \frac{\text{What you get}}{\text{What you pay}}$$

Different efficiencies

Faradaic efficiency (current efficiency)

- The completeness of the process.
- Do we get full conversion for all charge passed
- The extent to which Faraday's law is obeyed
- Normally high

$$\eta_{\mathrm{F}} = \frac{\dot{n}_{\mathrm{converted,real}}}{\frac{I}{nF}}$$

Voltage efficiency

- The ration voltage reclaimed during "discharge" and the voltage applied on "charge"
- Can be moderate or even low

$$\eta_{\mathrm{V}} = \frac{\int_0^t E_{\mathrm{out}}}{\int_0^t E_{\mathrm{in}}}$$

System efficiency

- The total energy efficiency of entire system including any balance of plant.

$$\eta_{\mathrm{sys}} = \frac{\mathrm{Desired\ energy}_{\mathrm{out}}}{\mathrm{Desired\ energy}_{\mathrm{in}}}$$

Just energy in and out?

$$\eta = \frac{\text{What you get}}{\text{What you pay}}$$

	Chargeable battery	Electrolyzer	Fuel cell
What you pay:	Electrical energy	Electrical energy	Chemical energy
What you get:	Electrical energy	Chemical energy	Electrical energy







Reference voltage for the energy of a fuel

$$E^0_{\text{cell}} = \frac{|\Delta g^0_{\text{r}}|}{nF} = 1.23\ \text{V}$$

Reversible cell voltage from Δg^0_r (HHV)

The energy of a fuel is not only the part freely availble for work (Δg^0_r).
It is the total energy (Δh^0_r).

$$E^0_{\text{TN}} = \frac{|\Delta h^0_{\text{r}}|}{nF} = 1.48\ \text{V}$$

Thermo-neutral voltage from Δh^0_r (HHV)
(*possible in electrolyzers*)
(*hypothetical in fuel cells*)

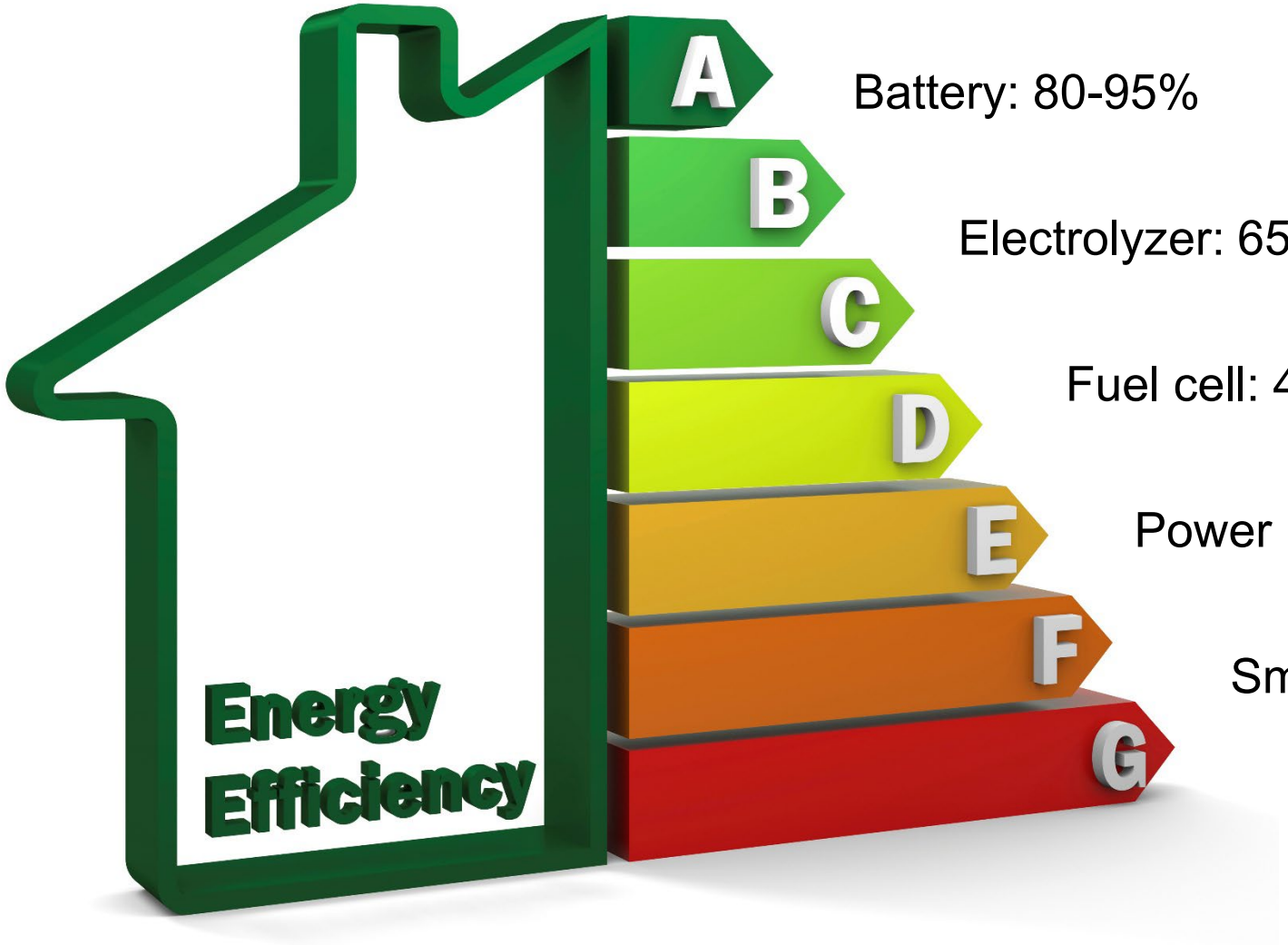
$$E_{\text{TN}} = \frac{|\Delta h_{\text{r}}|}{nF} = 1.25\ \text{V}$$

Thermo-neutral voltage from Δh_r (LHV)
(*possible in electrolyzers*)
(*hypothetical in fuel cells*)

The voltage efficiency (*fuel cells and electrolyzers*)

Real fuel cell (voltage) efficiency: $$\eta_{\mathrm{FC}} = \frac{E_{\mathrm{cell}}}{E_{\mathrm{TN}}}$$

Real electrolyzer (voltage) efficiency: $$\eta_{\mathrm{EC}} = \frac{E_{\mathrm{TN}}}{E_{\mathrm{cell}}}$$



Battery: 80-95%

Electrolyzer: 65-90%

Fuel cell: 40-60%

Power plant/Large marine ICE: 30-45%

Small combustion engine: 10-20%

Thank you for your attention